

Operational Readiness for LLM Hallucinations: A Counting-Process Capacity Model and Measured Evidence on Guardrail Cascades

Working draft. Revised from an earlier analytical-only draft; see Section 3.4 for corrections. Code and data: [hallucination_poisson/](#) in the accompanying repository.

Abstract

Organisations that deploy large language models (LLMs) must decide how much human triage capacity to hold in reserve and how many automated guardrail layers to place in front of the model. Practice relies on two rules of thumb: staff for the average incident load, and stack guardrails whose combined benefit is computed under independence. We model critical hallucinations as a (possibly over-dispersed, non-homogeneous) counting process, size capacity at a quantile of the daily count, and treat the multiplicative guardrail formula as the independence special case of Fréchet-type bounds. We show analytically that over-dispersion inflates required capacity by roughly the square root of the dispersion index, that positive association between layers makes the independence formula a lower bound on residual risk, and that, because capacity is integer-valued, total cost is not unimodal in the number of layers; an earlier claim that three layers are structurally optimal does not follow from the cost model. We then evaluate the integrated pipeline on measured outputs of three open LLMs answering 900 SQuAD 2.0 questions through three executed guardrail layers (rules, an extractive-QA classifier, an LLM judge) with measured latency and failures, with user traffic simulated as a non-homogeneous arrival process. A single validated classifier cut required capacity by 80-95%; a routed judge added a modest, model-dependent gain at about triple the latency. The independence formula understated the share of hallucinations passing a two-layer cascade by a factor of 1.6-2.6 (2.7-4.6 with routing), a classifier with chance-level discrimination appeared beneficial only because it blocked almost everything, and with mild burstiness (daily-rate CV 0.3) a Poisson-sized capacity plan was exceeded on about 25% of days instead of 5%. Arrival times are simulated, models are small, and the task is extractive QA, so the numbers do not transfer to production; the method and its diagnostics do.

Keywords: LLM operations, hallucination, guardrails, capacity planning, Poisson process, over-dispersion, design science

1. Introduction

Large language models (LLMs) are now embedded in customer-facing and decision-support information systems, where an ungrounded answer is a service incident rather than a quality blemish. IS research has examined adoption and value of generative AI, but the *operations* question that follows deployment has received little formal treatment: how much human triage and verification capacity must an organisation keep on standby, and how many automated guardrail layers should sit in front of the model, when hallucinations arrive at random?

Practitioners answer this with rules of thumb: staff for the average incident load, or stack as many guardrails as the latency budget tolerates. Both are costly mistakes. Staffing at the mean leaves the organisation under-provisioned on roughly one day in three to one in two, and the benefit of an additional guardrail is usually computed under an independence assumption that, as we show, is both unverified and optimistic.

We make three contributions, and we are explicit about which parts are new and which apply known results. **First**, we give an analytical capacity model for LLM incident handling in which critical hallucinations form a (possibly non-homogeneous, possibly over-dispersed) counting process, and the readiness requirement is the quantile of the daily count (Section 3). The quantile rule and the over-dispersion correction are classical (Whitt, 2007; Cameron and Trivedi, 1998); our contribution is to apply them to this arrival stream and to test whether the counting-process assumptions hold on measured LLM outputs. **Second**, we restate the multiplicative guardrail formula as the independence special case of Fréchet-type bounds on an arrival-rate (thinning) cascade (Proposition 3). Dependence between guardrail layers has been measured on real defence stacks (Alotaibi et al., 2026) and the independence product is known to over-credit redundancy (Bhardwaj et al., 2026); we add a measurement on hallucination-grounding layers and carry the correction through to the capacity decision. We then show that, once capacity is integer-valued, total cost is generally *not* unimodal in the number of layers and give comparative statics in the labour-to-compute cost ratio (Proposition 4); this follows from the model structure and is presented as a model property, not an empirical discovery. **Third**, we evaluate the integrated pipeline on real model outputs: three open models answering SQuAD 2.0 questions, three guardrail layers executed with measured latency and error behaviour, and traffic replayed through a non-homogeneous arrival process. To our knowledge, this is the first study that tests counting-process assumptions on measured LLM hallucination outputs and connects the result to quantile-based triage capacity and guardrail-depth decisions in one framework; each ingredient has precedents (Section 2). Arrival times are simulated; outcomes, guardrail decisions and latencies are measured.

The paper follows the design-science tradition in IS (Hevner et al., 2004; Gregor and Hevner, 2013): we propose an artifact (a capacity-planning and guardrail-sizing method), ground it in a formal model, and evaluate it on real system outputs. We are explicit about what the evaluation does and does not establish (Section 6).

2. Related Work

Hallucination and its mitigation. Surveys document hallucination as a pervasive property of neural text generation (Ji et al., 2023). Mitigation work includes retrieval grounding (Lewis et al., 2020), sampling-based consistency checks (Manakul et al., 2023), safeguard classifiers (Inan et al., 2023) and LLM-as-judge verification (Zheng et al., 2023). This literature evaluates *detectors* by accuracy; it does not ask how detector performance translates into the human capacity an organisation must hold in reserve.

Capacity planning under random demand. The use of the Poisson quantile to size service capacity is classical in service operations (Whitt, 2007), and time-varying demand is handled by piecewise-stationary staffing (Green et al., 2007). Empirical studies of call centres report that arrivals are well described by a Poisson process only conditional on a random, day-specific rate, i.e. they are over-dispersed (Brown et al., 2005); count-data econometrics offers the negative-binomial remedy (Cameron and Trivedi, 1998). We import these tools to a new arrival stream and test whether the same caveats apply.

Human review capacity for AI outputs. Queueing models of human review of AI output exist: Poisson escalation arrivals with Erlang-C staffing of human operators for agent escalations (Besanson, 2026), M/G/1 models of reviewing AI drafts (Bartolucci and Vivo, 2026), and fluid-limit routing between an LLM judge and human review (Lin and Zhang, 2026). These use mean-based stability or waiting-time criteria and

assume, rather than test, Poisson arrivals. We size capacity at a quantile of the incident-count distribution and test the arrival assumptions.

Counting-process models of AI failures. Public-health-style monitoring of AI incident counts has found strong over-dispersion relative to Poisson in incident-database counts of vehicle and deepfake incidents (Abraham et al., 2026), and Markov reliability models of LLM agents reduce to the Goel-Okumoto NHPP software-reliability model in the rare-failure regime (Tran-Truong and Le, 2026). Neither concerns hallucination or guardrail-bypass events, nor applies time-rescaling tests or capacity sizing.

Layered defences. The multiplicative formula for the residual rate of independent safeguards is standard in reliability engineering, and it over-credits redundancy when failures are positively associated (Esary et al., 1967). For LLM defence stacks, Alotaibi et al. (2026) measure positive failure correlation between seven defence layers (ϕ between 0.30 and 0.75 on JailbreakBench) and call the product a lower bound that no real stack attains; Bhardwaj et al. (2026) give a Fréchet-Hoeffding sandwich and measure high co-failure among same-model LLM components. Our dependence results should be read as additional evidence, in a different setting (hallucination-grounding layers), rather than as a first demonstration.

Gap. Prior work addresses these pieces separately: arrival models for AI incidents, human-review queueing, measured layer dependence, and guardrail cost-latency trade-offs. We integrate them for LLM hallucination operations and validate the integrated pipeline on measured outputs. The literature on LLM guardrails moved quickly in 2026; our search was not exhaustive and similar work may exist.

3. Model

3.1 Arrivals of critical hallucinations

Users send queries at times governed by an intensity $\Lambda(t)$ with a daily cycle. Each query independently produces a *critical hallucination* (a wrong or ungrounded answer that passes all guardrails and reaches the user) with probability q . By the thinning theorem (Kingman, 1993), critical hallucinations form a non-homogeneous Poisson process (NHPP) with intensity $q\Lambda(t)$. Let X_d be the number of events on day d , with mean $\lambda = \int_{\text{day}} \Lambda(t)dt$. Then $X_d \sim \text{Poisson}(\lambda)$ regardless of the within-day shape of Λ ; the within-day shape matters for *hourly* staffing, where the hour- h requirement is the quantile of $\text{Poisson}(\lambda_h)$.

Inter-arrival times are exponential only under a constant rate. Under the NHPP, the time-rescaled gaps $\Lambda(t_i) - \Lambda(t_{i-1})$ are i.i.d. $\text{Exp}(1)$ (the time-rescaling theorem), which gives a goodness-of-fit test that we use in Section 5. The memoryless corollary of the earlier draft of this work therefore applies to rescaled time: elapsed time since the last event carries no information, but time-of-day does.

If the daily rate is itself random, $\Lambda_d = \lambda M$ with $E[M] = 1$ and $\text{Var}(M) = c^2$, the daily count is a mixed Poisson variable with $\text{Var}(X_d) = \lambda + \lambda^2 c^2 = \lambda\phi$, where $\phi = 1 + \lambda c^2$ is the dispersion index. This is the Cox-process alternative to the pure Poisson null.

3.2 Readiness capacity

For reliability target α , the capacity requirement is the generalised inverse of the daily-count distribution:

$$k^*(\alpha, \lambda) = \min \{ k \in \mathbb{N}_0 : P(X_d \leq k) \geq \alpha \}$$

Proposition 1 (Poisson capacity). Let $X \sim \text{Poisson}(\lambda)$. (a) $k^*(\alpha, \lambda)$ is non-decreasing in λ . (b) $k^*(\alpha, \lambda) = 0$ if and only if $\lambda \leq -\ln \alpha$; for $\alpha = 0.95$ this threshold is 0.0513 events/day. (c) Provisioning at the mean, $k = \lfloor \lambda \rfloor$, leaves the system under-provisioned with probability $P(X > \lfloor \lambda \rfloor)$, which is at least 0.26 for every $\lambda \geq 1$, is about 0.35 at $\lambda = 3$ (not 0.42, which is $P(X \leq 2)$) and tends to 0.5 as λ grows. (d) $k^* = \lambda + z_{\alpha}\sqrt{\lambda} + O(1)$ as $\lambda \rightarrow \infty$.

Proof sketch. (a) $\text{Poisson}(\lambda)$ is stochastically increasing in λ . (b) $P(X = 0) = e^{-\lambda} \geq \alpha$ iff $\lambda \leq -\ln \alpha$. (c) On $[n, n+1)$ the probability $P(X > n)$ is increasing in λ , so its minimum over $\lambda \geq 1$ is $P(X > 1) = 0.264$ at $\lambda = 1$, and it is 0.353 at $\lambda = 3$ ($P(X \leq 3) = 0.647$). (d) follows from the Cornish-Fisher expansion of the Poisson quantile. ■

Proposition 2 (over-dispersion inflates capacity). Let $X_d | M \sim \text{Poisson}(\lambda M)$ with $E[M] = 1$ and $\text{Var}(M) = c^2 > 0$. Then (a) $\text{Var}(X_d) = \lambda\phi$ with $\phi = 1 + \lambda c^2$; (b) X_d dominates $\text{Poisson}(\lambda)$ in the convex order, so $E[(X_d - k)^+] \geq E[(Y - k)^+]$ for every k , where $Y \sim \text{Poisson}(\lambda)$; (c) $k^*_{\text{mixed}}(\alpha) = \lambda + z_{\alpha}\sqrt{\lambda\phi} + O(1)$, so the safety stock $z_{\alpha}\sqrt{\lambda}$ is inflated by the factor $\sqrt{\phi}$, and a plan sized with the Poisson quantile is exceeded on more than a $1 - \alpha$ share of days for large k .

Proof sketch. (a) is the law of total variance. (b) For Poisson, $\partial/\partial\lambda E[(Y - k)^+] = P(Y \geq k)$, which is non-decreasing in λ , hence $\lambda \mapsto E[(Y - k)^+]$ is convex; Jensen's inequality applied to the mixing variable M gives $E[(X_d - k)^+] = E[g(\lambda M)] \geq g(\lambda)$. (c) follows by normal approximation with variance $\lambda\phi$. ■

Table 1 evaluates Proposition 2 exactly (negative-binomial mixing). At $\lambda = 10$ events/day and $c = 0.3$, the Poisson plan $k^* = 15$ is exceeded on 11.0% of days instead of 4.9%, and the correct requirement is 18; at $\lambda = 25$ and $c = 0.3$ the Poisson plan of 33 is exceeded on 16.7% of days and 41 is required.

Table 1. Capacity under over-dispersed arrivals (Gamma-mixed Poisson, $\alpha = 0.95$)

λ (events/day)	c (CV of daily rate)	ϕ	k^* Poisson	k^* mixed	P(exceed) Poisson plan
10	0	1.00	15	15	4.9%
10	0.1	1.10	15	16	5.7%
10	0.2	1.40	15	17	8.0%
10	0.3	1.90	15	18	11.0%
10	0.5	3.50	15	21	16.4%
25	0	1.00	33	33	5.0%
25	0.1	1.25	33	35	7.0%
25	0.2	2.00	33	37	11.9%
25	0.3	3.25	33	41	16.7%
25	0.5	7.25	33	50	23.0%

Exact negative-binomial quantiles; computed by paper_tables.py.

3.3 Guardrail cascades and dependence

A cascade of N layers with stand-alone clearance rates c_i (the share of critical hallucinations layer i blocks) is usually assumed to give $\lambda_N = \lambda_0 \prod (1 - c_i)$. This presumes independent layer decisions. In practice the same ambiguous or adversarial outputs tend to fool all layers.

Proposition 3 (dependence bounds). For any joint behaviour of the layers, $\lambda_0(1 - \sum c_i)^+ \leq \lambda_N \leq \lambda_0(1 - \max_i c_i)$. If the layers' catch indicators, conditional on a hallucination, are positively associated (Esary et al., 1967), then additionally $\lambda_N \geq \lambda_0 \prod (1 - c_i)$. Consequently the

multiplicative formula under-states residual risk and, through Proposition 1(a), under-states required capacity.

Proof sketch. The bounds are the Fréchet-Bonferroni bounds for the event that no layer fires. For associated indicators, $P(\cap \{I_i = 0\}) \geq \Pi P(I_i = 0)$ (Esary et al., 1967). ■

Table 2 illustrates the width of the identification problem with $c_i = 0.63$ and $\lambda_0 = 10$: without knowing the dependence, three layers could imply anything from 0 to 7 units of capacity, which is why the dependence must be *measured* (Section 5.2).

Table 2. Residual rate and capacity bounds ($\lambda_0 = 10$, $c_i = 0.63$, $\alpha = 0.95$)

N	λ_N lower	λ_N independent	λ_N upper	k* lower	k* indep.	k* upper
1	3.70	3.70	3.70	7	7	7
2	0.00	1.37	3.70	0	4	7
3	0.00	0.51	3.70	0	2	7
4	0.00	0.19	3.70	0	1	7

3.4 Cost-minimising guardrail depth

Let C_h be the daily cost of one unit of triage capacity, g_i the daily compute cost of layer i , $k_{\min} \geq 0$ a minimum standing capacity (an on-call engineer is not removed because incidents are rare), and consider the cost

$$f(N) = C_h \cdot \max\{k_{\min}, k^*(\alpha, \lambda_N)\} + \sum_{i \leq N} g_i \quad (+ \text{ latency-feasibility: } \sum L_i \leq SLA).$$

Proposition 4 (optimal depth). (a) f is in general not unimodal in N : because k^* is integer-valued with plateaus, f can rise and then fall again. (b) With $k_{\min} \geq 1$, let N_0 be the first N with $k^*(\alpha, \lambda_N) \leq k_{\min}$; every $N > N_0$ is strictly dominated, so $N \leq N_0$. (c) Let $\theta = C_h / g$ with equal per-layer compute cost g . The minimiser $N(\theta)$ is non-decreasing in θ : when human triage is relatively expensive, deeper cascades are optimal (Topkis, 1978). (d) The latency constraint restricts N to a feasible prefix of the cascade.

Proof sketch. (a) is shown by the example in Table 3. (b) For $N > N_0$, labour cost is constant at $C_h k_{\min}$ while compute increases. (c) Write $f(N) = g[N + \theta K(N)]$ with K non-increasing; the difference $f(N+1) - f(N) = g[1 + \theta(K(N+1) - K(N))]$ is non-increasing in θ , so the largest minimiser is non-decreasing in θ by Topkis' monotone comparative statics. (d) is immediate. ■

Correction to the earlier draft. The earlier version of this paper reported Table 3 up to $N = 5$ and concluded that three layers are "optimal" with a cost rebound at $N = 5$. Extending the same table to $N = 6$ and 7 shows the cost falls again at $N = 6$ when k^* reaches zero ($\lambda_6 = 0.026 < -\ln 0.95$), so under $k_{\min} = 0$ the minimum is $N = 6$, and the table itself ranks $N = 4$ (\$250) below $N = 3$ (\$375). With a realistic floor of one on-call engineer ($k_{\min} = 1$) the minimum is $N = 4$ (\$250). The claim that three layers are structurally optimal is not supported by the model; the optimal depth is an outcome of the cost ratio, the clearance rates and the dependence structure.

Table 3. Daily cost by guardrail depth ($\lambda_0 = 10$, $c = 0.63$ independent, $C_h = \$150/\text{unit}$, $g = \$25/\text{layer}$)

N	λ_N	k*	Labour	Compute	Total ($k_{\min} = 0$)	Total ($k_{\min} = 1$)
0	10.000	15	\$2,250	\$0	\$2,250	\$2,250

N	λ_N	k^*	Labour	Compute	Total ($k_{\min} = 0$)	Total ($k_{\min} = 1$)
1	3.700	7	\$1,050	\$25	\$1,075	\$1,075
2	1.369	4	\$600	\$50	\$650	\$650
3	0.507	2	\$300	\$75	\$375	\$375
4	0.187	1	\$150	\$100	\$250	\$250
5	0.069	1	\$150	\$125	\$275	\$275
6	0.026	0	\$0	\$150	\$150	\$300
7	0.009	0	\$0	\$175	\$175	\$325

Minimum: $N = 6$ under $k_{\min} = 0$; $N = 4$ under $k_{\min} = 1$. Costs are the earlier draft's illustrative assumptions.

4. Empirical Design

4.1 Data, models and ground truth

We use the SQuAD 2.0 validation set (Rajpurkar et al., 2018), which combines answerable questions with questions that the supplied passage cannot answer. A random sample of 300 questions (seed 0) is drawn for each model. Each model receives the passage and the question and must reply with a short answer or the token UNANSWERABLE. A response is labelled a **critical hallucination** if it is not an abstention and either (i) the question is answerable and the answer matches no gold answer (normalised containment or token-F1 ≥ 0.6) or (ii) the question is unanswerable. Abstentions are never counted as hallucinations and bypass the guardrails. The generators are Qwen2.5-0.5B-Instruct, SmolLM2-360M-Instruct and Qwen2.5-1.5B-Instruct (Qwen Team, 2024; Allal et al., 2025), run greedily on CPU. We use small open models because they can be run end-to-end with full control and measured latency; their hallucination rate is higher than that of frontier models, so absolute rates should not be transferred (Section 6).

4.2 Guardrail layers

The three layers follow the asymmetric architecture of the earlier draft, but each is implemented and executed on every answer:

- **L1, deterministic.** Regular expressions for e-mail, SSN, card and phone patterns; template-token leakage; degenerate repetition; and a grounding heuristic that blocks answers containing a number absent from the evidence passage.
- **L2, classifier.** A small NLI cross-encoder (DeBERTa-v3-small; He et al., 2021) scoring $P(\text{passage entails "the answer to Q is A"})$; the output is blocked if this probability is below 0.5. (The earlier draft named Llama Guard-type models for this layer; these target safety policy violations, not factual grounding, so an entailment classifier is the appropriate classifier for hallucinations.)
- **L3, semantic judge.** An LLM judge (Qwen2.5-1.5B-Instruct) given the passage, question and answer, replying SUPPORTED or UNSUPPORTED (Zheng et al., 2023). It is invoked only when routed: when L2 is present, only if the L2 entailment probability is below 0.9.

All layers use the same evidence passage as retrieval-grounded verification. Every layer is run on every example so that any subset or cascade can be evaluated offline from observed flags, without any independence assumption. We record flag, score, wall-clock latency and exceptions per call. Layers fail open; a call that errors or exceeds a 2 s timeout is counted as a system failure, from which we estimate $\lambda_{\text{sys_fail}}$ rather than assuming it.

4.3 Traffic replay and estimators

Production logs are unavailable, so *arrivals* are simulated while *outcomes* are measured. Queries arrive according to an NHPP with a business-hours-shaped 24-slot intensity and a mean of 200 queries/day for 90 days (a Cox variant multiplies each day's intensity by a $\text{Gamma}(1, c^2)$ factor, $c = 0.3$, to represent bursty demand). Each arriving query replays a uniformly drawn example from the measured pool; it becomes a critical event if it is a hallucination that survives the cascade. For each resulting event stream we estimate λ = events/days and test: (i) the index-of-dispersion test on daily counts; (ii) a chi-square test of constant hourly intensity; (iii) a Kolmogorov-Smirnov test of raw inter-arrival times against an exponential, and of time-rescaled gaps under the fitted hourly intensity (the Lilliefors-type p-value is optimistic); (iv) a negative-binomial fit of daily counts. Uncertainty from the finite evaluation pool is quantified by a 300-draw bootstrap over examples (95% percentile intervals).

Because the same pool generates all arrivals, the replay tests whether the *counting process* has the claimed structure given measured per-query outcomes; it does not test whether real users' queries are independent. That is the main external-validity limit (Section 6).

4.4 Cost model

Daily cost is $C_h \cdot \max(k_{\min}, k^*) + \text{compute} + \text{false-block cost} + \text{residual-harm cost}$. We set $C_h = \$150$ per capacity unit (as in the earlier draft), compute \$0.0001 per classifier call and \$0.002 per judge call, and treat the false-block cost c_{fp} and residual-harm cost c_{res} as scenario parameters in a sensitivity grid. These are assumptions, not estimates, and are labelled as such wherever used.

5. Results

5.1 Hallucination rates and guardrail layer performance

On 300 SQuAD 2.0 questions per model, the critical-hallucination rate is 60.3% (Qwen2.5-0.5B), 65.0% (SmolLM2-360M) and 47.0% (Qwen2.5-1.5B). Most hallucinations are answers to unanswerable questions. Table 4 reports the layers. **Layer L1** (rules) catches 0.7-5.1% of hallucinations: this task contains no PII or template leaks and few fabricated numbers, so the deterministic layer is nearly inert here, though free of latency and false blocks. **Layer L3** (the 1.5B judge) blocks 63-84% of hallucinations but also 25-39% of correct answers, and each call takes about 1.9 s on our CPU, with 25-30% of calls exceeding the 2 s timeout.

Negative result for the NLI classifier. Our first implementation of L2 was the NLI cross-encoder of the earlier draft. It was essentially uninformative: its entailment score separates correct from hallucinated answers with AUC 0.52-0.58 (0.5 is chance), and at the default threshold it blocked 87-88% of hallucinations and 85-90% of *correct* answers. Its apparent capacity savings were therefore an artefact of blocking nearly everything. We replaced it by an extractive-QA verifier (a SQuAD 2.0-trained RoBERTa model whose own answer span is compared with the generator's answer by token-F1; AUC 0.84-0.90) and re-ran all layers on the stored answers. This verifier is specialised to the task family (it was trained on SQuAD), which flatters it relative to a generic classifier; all numbers below use it, and we keep the NLI results to show that a "classifier layer" is not automatically a useful one.

Table 4. Guardrail layers on 300 SQuAD 2.0 questions per generator (CPU inference, latency re-measured without concurrent jobs)

Generator	p_0 (hallucination rate)	c_1 (L1)	c_2 (L2 QA)	c_3 (L3 judge)	L2 false-block (good answers)	L3 false-block	L2 mean ms	L3 mean ms
Qwen2.5-0.5B	60.3%	2.8%	89.5%	84.0%	15.1%	39.5%	98	1941
SmolLM2-360M	65.0%	5.1%	96.9%	63.1%	69.5%	24.8%	97	1946
Qwen2.5-1.5B	47.0%	0.7%	81.6%	78.7%	13.2%	33.3%	95	1898

c_i = stand-alone share of hallucinated answers the layer blocks. False-block = share of correct (non-hallucinated, non-abstained) answers blocked. The L2 threshold ($F1 < 0.5$) was not tuned; SmolLM2 gives longer answers, which lowers its token-F1 agreement and inflates its L2 false-block rate.

5.2 Dependence between layers: the independence formula under-states residual risk

Among hallucinated answers, the catch decisions of L2 and L3 are positively correlated ($\phi = 0.15$ - 0.24 ; L1 pairs: $\phi \leq 0.09$), and the observed joint catch rate exceeds the independence prediction in every pair for every model, in the direction of Proposition 3. The consequence for the residual rate is large (Figure 2, left). For the L2+L3 cascade, the share of hallucinations that pass all layers is 1.1-3.9% under the independence formula, but 2.6-6.4% when both layers are run on every answer (1.6-2.6 times higher), and 3.1-15.6% with dynamic routing of only low-agreement answers to the judge (2.7-4.6 times higher). Routing therefore has a first-order cost that independence-based sizing hides: it reduces judge calls by about 89%, but the judge no longer sees answers that the verifier considers consistent, and those contain a substantial share of the hallucinations.

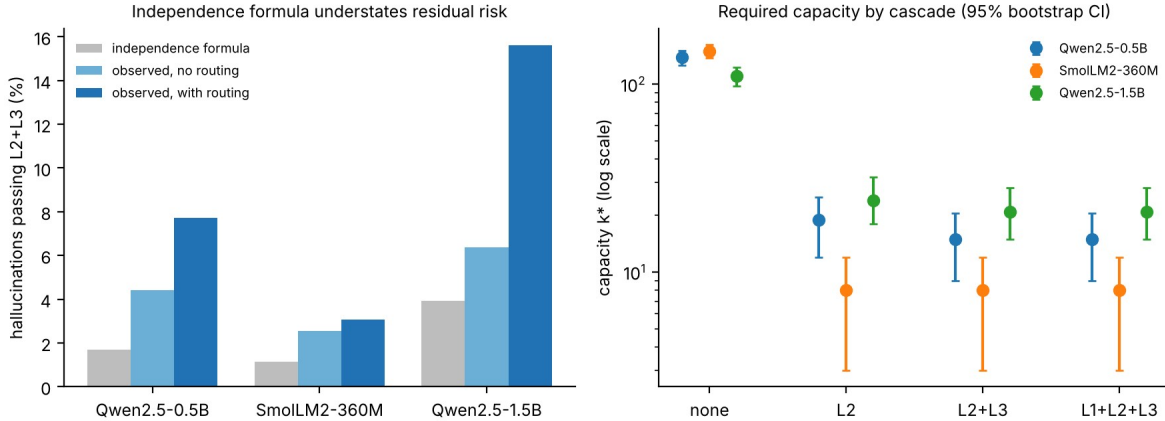


Figure 2. Left: share of hallucinations passing the L2+L3 cascade under independence, observed without routing, and observed with routing. Right: capacity k^* by cascade (95% bootstrap CI over the evaluated pool); log scale.

5.3 Counting-process checks under replayed traffic

With non-homogeneous Poisson arrivals and no burstiness, the replay reproduces the Poisson structure, as it must by construction: for the unprotected stream, the daily-count dispersion test does not reject ($p = 0.53$ - 0.91), the raw inter-arrival times are strongly non-exponential ($p < 10^{-100}$) because the intensity varies with the hour (hour-of-day homogeneity rejected), and the *time-rescaled* gaps are consistent with $\text{Exp}(1)$ ($p = 0.50$ - 0.85). We stress that this is a check that the estimators and the replay are correct and that the NHPP/rescaling machinery behaves as theory predicts; it is not evidence that real user traffic is a Poisson

process, because the arrivals are simulated (Section 4.3). Within-day non-homogeneity matters for staffing by hour but not for the daily count.

5.4 Burstiness: Poisson capacity plans fail

When the daily intensity is multiplied by a Gamma factor with a coefficient of variation of only 0.3, the daily-count dispersion test rejects overwhelmingly ($p < 10^{-90}$ for the unprotected stream) and the negative-binomial fit gives r between 12.7 and 13.4. The Poisson capacity $k^* = 136, 146, 108$ (for the three generators, no guardrails) is then exceeded on 23-28% of days instead of the targeted 5%; the correct negative-binomial requirement is 178, 194, 141, i.e. 31-33% more (Figure 1; Proposition 2). Under the best cascade (L1+L2, $\lambda \approx 4$ -17/day) the effect is smaller in absolute capacity but still raises the exceedance probability of the Poisson plan to 7-10%. The value $c = 0.3$ is a scenario, not an estimate: the point is that a mild day-to-day variation in demand, which any real service has, is enough to invalidate the Poisson plan, so the dispersion must be estimated from production incident logs before capacity is fixed.

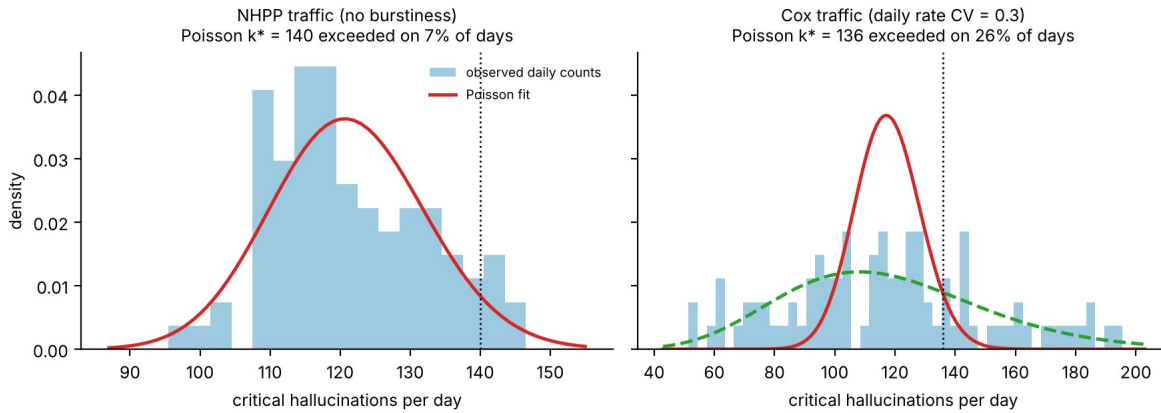


Figure 1. Daily critical hallucinations without guardrails (Qwen2.5-0.5B, 90 days, 200 queries/day). Left: steady NHPP traffic; right: bursty traffic. Dotted line: Poisson k^* ($\alpha = 0.95$).

5.5 Capacity, latency and cost by cascade

Table 5 gives, for each generator, the residual rate λ , the capacity k^* , the added latency and the false-block volume of the main cascades at 200 queries/day, with bootstrap intervals over the evaluated pool. Intervals are wide at this pool size (300 questions), so differences of a few units of k^* between cascades are not significant; the large differences are.

Table 5. Cascade performance at 200 queries/day, $\alpha = 0.95$, SLA 500 ms mean added latency (95% bootstrap CI in brackets)

Generator	Cascade	λ per day	k^*	Mean added latency (ms)	False blocks / day	System failures / day
Qwen2.5-0.5B	none	120.7 [109, 132]	139 [126, 151]	0	0.0	0.0
Qwen2.5-0.5B	L2	12.7 [7, 18]	19 [12, 25]	98	12.0	0.0
Qwen2.5-0.5B	L3	19.3 [12, 25]	27 [18, 34]	1941 ✗	31.3	60.0
Qwen2.5-0.5B	L2+L3	9.3 [5, 14]	15 [9, 21]	311	17.3	6.0
SmolLM2-360M	none	130.0 [120, 142]	149 [138, 162]	0	0.0	0.0
SmolLM2-360M	L2	4.0 [1, 7]	8 [3, 12]	97	48.7	0.0
SmolLM2-360M	L3	48.0 [38, 57]	60 [48, 70]	1933 ✗	17.3	50.0

Generator	Cascade	λ per day	k^*	Mean added latency (ms)	False blocks / day	System failures / day
SmolLM2-360M	L2+L3	4.0 [1, 7]	8 [3, 12]	271	50.7	6.7
Qwen2.5-1.5B	none	94.0 [82, 106]	110 [97, 123]	0	0.0	0.0
Qwen2.5-1.5B	L2	17.3 [12, 24]	24 [18, 32]	88	14.0	0.0
Qwen2.5-1.5B	L3	20.0 [13, 26]	28 [19, 35]	1752 ✗	35.3	49.3
Qwen2.5-1.5B	L2+L3	14.7 [9, 21]	21 [15, 28]	296	22.0	6.7

✗ = violates the 500 ms budget. System failure = a layer call that errors or exceeds 2 s. $L1+L2$ and $L1+L2+L3$ equal $L2$ and $L2+L3$ up to small differences because $L1$ is nearly inert on this task.

Capacity. A single well-chosen classifier reduces required capacity from 139, 149, 110 to 19, 8, 24 units/day (a 78-95% reduction). Adding the routed judge reduces it further only modestly (15, 8, 21) and for SmolLM2 not at all, while multiplying added latency by about three and adding roughly 6 system failures per day through judge timeouts. Running the judge on every answer without routing is infeasible under any latency budget we consider (about 1.9 s) and produces about 50-60 system failures per day on our CPU setup.

Latency budget. Under the 100 ms budget of the earlier draft, only $L1$ and $L2$ ($L2$ takes about 90-100 ms here) are feasible and no judge layer fits; under 500 ms the routed judge fits on average but not at the 95th percentile (about 1.5 s). Latencies are CPU-bound and would be much lower on GPUs or hosted inference.

Cost. With $C_h = \$150$ per capacity unit, \$1 per falsely blocked answer and \$100 per hallucination reaching a user (illustrative scenario values), the cheapest cascade is $L1+L2+L3$ for Qwen2.5-0.5B, $L1+L2$ for SmolLM2-360M, $L1+L2+L3$ for Qwen2.5-1.5B; daily cost falls from \$32,917, \$35,350, \$25,900 without guardrails to \$3,201, \$1,649, \$4,639. The ranking is the same across the traffic $\times$ false-block-cost $\times$ harm-cost grid of Section 4.4 (27 scenarios per generator), but the dollar levels are assumptions. Whether the third layer pays for itself depends on the generator: it does for Qwen2.5-0.5B and Qwen2.5-1.5B and does not for SmolLM2-360M, so a universal "three layers" prescription is not supported by these data. The classifier is the dominant layer, the deterministic layer is nearly free but also nearly useless on this task, and the judge's marginal value is positive but small and uncertain.

6. Discussion

6.1 Implications for IS practice and research

- **Size capacity from a quantile, and fit the dispersion first.** Provisioning at the mean load fails on roughly 40-50% of days in every configuration we measured. The Poisson quantile fixes this when daily counts are Poisson, but Proposition 2 and the burst experiment show that a modest day-to-day variation in demand (CV 0.3) multiplies the miss rate several-fold. Dispersion should be estimated from incident logs before the plan is fixed.
- **Guardrail value is an empirical, per-deployment quantity.** The layers' standalone clearance, their dependence, their false-block cost and their latency vary strongly by layer and model. In our data one well-chosen classifier delivers most of the benefit, the deterministic layer is nearly inert, and the judge adds a modest, model-dependent gain at triple the latency. A fixed "three layers" prescription is not supported; the method here selects the stack from measurements. A classifier that is not validated can be actively harmful: our NLI layer had chance-level discrimination and appeared to cut capacity only because it blocked almost everything.

- **Routing is a first-order design choice.** Sending only low-agreement answers to the expensive judge cuts judge calls by roughly an order of magnitude, but it also forgoes catches among answers the classifier rated as consistent; routing raised the residual hallucination share by a factor of 1.2-2.4 relative to running both layers on everything (Section 5.2).
- **Governance.** Because k^* converts a detector's precision/recall into headcount, the framework gives IT governance a common unit (units of triage capacity or dollars per day) in which to compare detectors, set the SLA and justify standing on-call capacity.

6.2 Limitations

- **Simulated arrivals.** Outcomes are measured on real model outputs, but arrival times are simulated and each arrival replays a bootstrapped example. Real users' queries are correlated (trending topics, incidents), so real streams are likely *more* over-dispersed than the pure replay. The burst experiment bounds this effect but does not replace production data; validation on proprietary logs is the main follow-up.
- **Small open models and one task.** We study 0.4-1.5B-parameter models on extractive QA with unanswerable questions. Absolute hallucination rates are much higher than for frontier models, and the layers are CPU-bound; latencies on GPU or hosted APIs differ. The *method* transfers, the *numbers* do not.
- **Label noise.** Ground truth uses string and token-F1 matching against SQuAD gold answers, which can mark valid paraphrases as hallucinations; this affects all configurations alike but inflates the base rate.
- **Cost parameters are assumptions.** Only labour-to-compute ratios and the sensitivity grid are informative; dollar levels are illustrative.
- **Single judge and classifier.** Layer dependence and clearance were measured for one NLI model and one judge model; other choices may be more or less correlated.
- **Independent days.** The model treats days as exchangeable; weekly seasonality and trends are not modelled.

7. Conclusion

Treating critical hallucinations as a counting process turns guardrail performance into a staffing and cost decision. The analysis yields three results that differ from the common practice the earlier draft followed: capacity must be sized from an over-dispersion-aware quantile; the multiplicative guardrail formula is an optimistic bound whose gap must be measured; and the cost-minimising number of layers is not fixed at three but depends on integer capacity effects, the floor on standing staff, latency, the labour-to-compute ratio and, empirically, on the generator. On measured outputs of three open models, a single task-validated classifier reduces the required capacity by roughly 80-95% relative to no guardrails; a routed judge adds a further modest, model-dependent reduction at about triple the latency, and every cascade incurs false blocks and system failures that the earlier framework ignored. The independence formula understated the share of hallucinations passing a two-layer cascade by a factor of 1.6-2.6 (2.7-4.6 with routing), and a mild burstiness (daily-rate CV of 0.3) raised the exceedance rate of a Poisson-sized plan from 5% to about 25%. Validating the counting-process assumptions on production incident logs is the key next step.

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